CSCI 788 — PROJECT SPECIFICATION

Grounded Multi-Agent RAG for Healthcare Workflow Automation

Verifiable, policy-grounded decision support for multi-agent LLM systems

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WHY THIS MATTERS

Bigger Picture & Connections

Big Picture

AI is entering high-stakes healthcare decisions

Prior authorization, claims review, and care management are being automated with LLMs. A wrong or unsupported decision doesn't just cost accuracy — it can delay patient care or trigger compliance failures.

Connected Problem

Multi-agent RAG pipelines lack verifiable grounding

Multi-agent systems split reasoning across specialized agents, but most implementations don't verify that each agent's output is actually traceable to the retrieved source document — grounding is assumed, not measured.

THE GOAL

What This Project Is About

Build a multi-agent RAG system that produces healthcare workflow decisions where every claim is explicitly traceable to a retrieved source document.

Illustration — a prior-authorization request



What Has Already Been Done

Retrieval-Augmented Generation

Grounding LLM outputs in retrieved documents to reduce reliance on parametric memory alone.

Multi-Agent LLM Orchestration

Frameworks such as LangGraph and AutoGen decompose complex reasoning across cooperating agents.

Hallucination Auditing

Growing body of work measuring and stress-testing how well LLMs stay faithful to source material under pressure.

Healthcare NLP & Clinical Decision Support

Domain-specific work on applying language models to clinical and administrative healthcare tasks.

The gap this project addresses

Few systems combine agent-level task decomposition with explicit, measured grounding evaluation in a healthcare workflow setting — most either optimize for task completion or for factuality, not both together.

ATTACK PLAN & MILESTONES

Fall Semester Timeline

Sept

Finalize scope, literature review, design synthetic case set

Early Oct

Build RAG ingestion pipeline and single-agent baseline

Late Oct

Implement multi-agent orchestration in LangGraph

Early Nov

Build evaluation framework and groundedness metrics

Mid Nov

Run experiments, compare against baseline, analyze results

Early Dec

Final report, poster, and presentation

EXPECTED RESULTS & DELIVERABLES

What Success Looks Like

Final deliverable

An end-to-end multi-agent RAG system (LangGraph) producing structured, cited decisions across eligibility, medical necessity, and documentation checks.

Data sources

Policy documents: publicly available CMS coverage-determination and prior-authorization guidelines. **Case scenarios:** synthetically generated, matched against those policies — no real patient or employer data used.

Evaluated across four dimensions on a held-out synthetic case set with known correct outcomes:

Groundedness

% of output claims traceable to a retrieved source passage

Hallucination Rate

% of claims with no supporting evidence in retrieved text

Decision Accuracy

Agreement with known-correct outcomes on held-out cases

Latency

End-to-end response time vs. a single-agent baseline

Thank You

Questions & Discussion

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